K J Somaiya Institute of Technology

An Autonomous Institute Permanently Affiliated to University of Mumbai.

A Project Based Learning

Report On



“Indian Meteorological Department Aviation Forecast Verification Web Portal with ML-Based Aerodrome Warning Prediction”

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Guide

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CERTIFICATE

This is to certify that the project titled "**Indian Meteorological Department Aviation Forecast Verification Web Portal with ML-Based Aerodrome Warning Prediction**" is completed under my supervision and guidance in partial fulfilment of the requirements of the course Project Based Learning - Minor Project Lab-1, by the following students:

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The course is a part of semester V of the Department of Artificial Intelligence and Data Science during the academic year 2026-2027. The said work has been assessed and is found to be satisfactory.

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1. Introduction

1.1 Aviation Meteorology and Flight Safety

Aviation meteorology is the application of atmospheric science to the planning and safe conduct of air operations. Parameters such as surface wind, gusts, visibility, cloud, temperature, pressure (QNH) and significant weather such as thunderstorms directly influence take-off, climb, approach and landing. Internationally, the provision of meteorological service to air navigation is governed by ICAO Annex 3 [2], while the codes used to exchange observations and forecasts, including METAR, are defined in WMO Manual on Codes No. 306 [1]. Accurate and timely forecasts therefore form a critical input to flight planning and operational decision-making.

1.2 Indian Meteorological Department

The India Meteorological Department (IMD) is the national agency responsible for meteorological observations, weather forecasting and warnings in India [3]. Its aviation meteorological offices issue products such as the Take-Off Forecast (TOF), Local Area Forecast (LAF) and Aerodrome Warning (ADWR) for airports. The quality of these products is assessed periodically by comparing them against what was actually observed, a process known as forecast verification.

1.3 Problem with Manual Verification

As stated in the project problem statement, the aviation forecast verification process at IMD is largely manual, which makes it time-consuming, error-prone and difficult to scale. Forecast files and observation reports are available in different formats (text, PDF and CSV) and must be matched in time and compared parameter by parameter. The absence of an automated system delays performance evaluation and limits the feedback available for improving forecast accuracy.

1.4 The Aviation Forecast Verification and Prediction Portal

The project was developed in coordination with senior IMD officials so as to remain aligned with operational requirements. It is a single web-based portal that automates the verification of TOF, LAF and ADWR by comparing forecasts with METAR observations (for TOF and ADWR) and upper-air observations (for LAF), and generates accuracy tables, graphs and downloadable reports. The same portal also includes a Machine Learning-based Aerodrome Warning Prediction component: two logistic regression classifiers estimate the probability of TSRA (thunderstorm with rain) in the next METAR observation and of a gust of 25 knots or more, and a linear regression model estimates wind speed, wind gust and wind direction. The application has been hosted on IMD infrastructure and is currently in the testing and validation phase.

1.5 Forecast Verification and Forecast Prediction

The two functions of the portal address the same operational need from two directions. Forecast verification is retrospective: an already-issued forecast is compared with observations to measure how good it was. Forecast prediction is prospective: historical METAR observations are used to estimate what is likely to occur next. Both rely on the same observation data, and together they let forecasters review past performance and consider a data-driven estimate of upcoming hazards.

2. Aim & Objective

2.1 Aim

To develop a web-based portal that automates the verification of aviation weather forecasts issued by IMD against actual meteorological observations, and that supports meteorological personnel with data-driven insights on aerodrome-warning-relevant weather through machine learning-based prediction.

2.2 Objectives

- To automate the verification of Take-Off Forecasts (TOF), Local Area Forecasts (LAF) and Aerodrome Warnings (ADWR).
- To compare forecast data with observed METAR and upper-air observations using standard meteorological verification criteria.
- To generate verification reports, accuracy tables, graphs and performance metrics.
- To predict the occurrence of TSRA in the next METAR observation and gust events of 25 knots or above.
- To estimate wind speed, wind gust and wind direction using linear regression.
- To support forecasters in improving the reliability and accuracy of aviation weather forecasts, with the prediction output serving as decision support and not as an official forecast.

3. Literature Survey

This survey covers the standards and published work that underpin the project. Only sources that can be verified are cited; full details are given in Chapter 10.

3.1 Aviation Meteorology and Forecasting Standards

ICAO Annex 3 specifies the meteorological service to be provided to international air navigation, including aerodrome forecasts, aerodrome warnings and the observations on which they depend [2]. IMD, as the national meteorological authority, provides these services in India [3]. Murphy's essay on forecast goodness argues that a forecast should be judged on consistency, quality and value, which motivates systematic verification of operational forecasts [10].

3.2 METAR Observations and Aviation Weather Codes

METAR is the routine aerodrome weather report whose format and code tables are defined in WMO Manual on Codes No. 306 [1]. A METAR reports wind direction and speed (with gusts), visibility, present weather (for example TS, RA, TSRA), cloud, temperature, dew point and QNH. METAR archives are available from services such as OGIMET [4], and upper-air sounding data from the University of Wyoming archive [5]; these are the observation sources named in this project.

3.3 Forecast Verification Methodologies

Jolliffe and Stephenson describe the standard framework of forecast verification: matching forecasts to observations, forming contingency tables for categorical (yes/no) events, and computing scores such as accuracy, hit rate and false-alarm measures, while continuous variables are assessed by error measures [8]. Murphy and Winkler's distributions-oriented view and the WMO/WWRP verification working group resources provide further guidance [6][7]. The portal applies tolerance-based criteria (for example ±30° wind direction) as used by IMD, which is a form of categorical hit/miss verification per parameter.

3.4 Meteorological Warning Systems

Aerodrome warnings alert aerodrome users to hazards such as thunderstorms and strong winds [2]. Verifying them requires checking whether the warned phenomenon occurred in the validity period, which is an occurrence/non-occurrence problem and is the basis of the ADWR verification module.

3.5 Machine Learning in Weather Prediction

Statistical post-processing has long been used in meteorology; Glahn and Lowry introduced Model Output Statistics using regression to relate predictors to weather elements [11]. McGovern et al. review the use of artificial intelligence to support real-time decisions for high-impact

weather and stress that such tools should aid forecasters [12]. Wilks provides the statistical foundation for applying regression and probabilistic methods to atmospheric data [9].

3.6 Logistic and Linear Regression

Logistic regression models the probability of a binary outcome through the logistic function and is well established for event/non-event problems [13][14]; it suits TSRA and gust-exceedance prediction. Linear regression models a continuous response as a weighted sum of predictors [14] and suits estimation of wind speed, gust and direction. For threshold selection and evaluation of classifiers, precision, recall and the F1 score are standard, and ROC analysis is a common companion technique [15].

3.7 Research Gap

Existing material largely treats verification and prediction separately. This project places both in a single operational portal: automated verification of issued products, and a supporting predictive module trained on historical METAR data.

4. Social Impact/Effect of Problem

Aviation safety and passenger welfare. Weather is a recognised factor in aviation operations, and the reliability of aviation weather products affects the confidence with which operational decisions are taken. Improved forecast quality ultimately benefits passengers, crew and airport users.

Impact of thunderstorms and gusts. Thunderstorms can bring heavy rain, lightning, turbulence and sudden wind shifts, while strong gusts affect take-off and landing, particularly for light aircraft. The warning of such hazards in aerodrome warnings is therefore of operational importance, and so is knowing how reliable those warnings are.

Importance of timely warnings. A warning is valuable only if it is issued in time and proves reliable. By quantifying warning accuracy (for the sample shown in Chapter 8, thunderstorm warnings and gust/wind warnings are scored separately), the portal highlights where warnings need to improve.

Reduced workload and consistency. Automation removes repetitive manual comparison from meteorological personnel, reduces transcription errors, and applies the same criteria every time, giving consistent and faster forecast evaluation.

Contribution of ML insights. The ML module offers probabilistic estimates of TSRA and gusts that forecasters may consider alongside their own analysis. It is a decision-support aid only. The system does not control flights, does not issue official warnings and does not guarantee flight safety.

5. System Design

5.1 Overall System Architecture

The portal follows a pipeline architecture. A user selects an ICAO station and a UTC time period; the portal accepts the relevant forecast files (TOF text, LAF PDF, ADWR text) and observation data (METAR reports, upper-air CSV); the Verification Engine performs time matching, parameter comparison and accuracy calculation; and results are presented as accuracy percentages, tables, graphs and downloadable reports (Fig. 5.1). The ML prediction module is added alongside this engine and uses historical METAR observations.

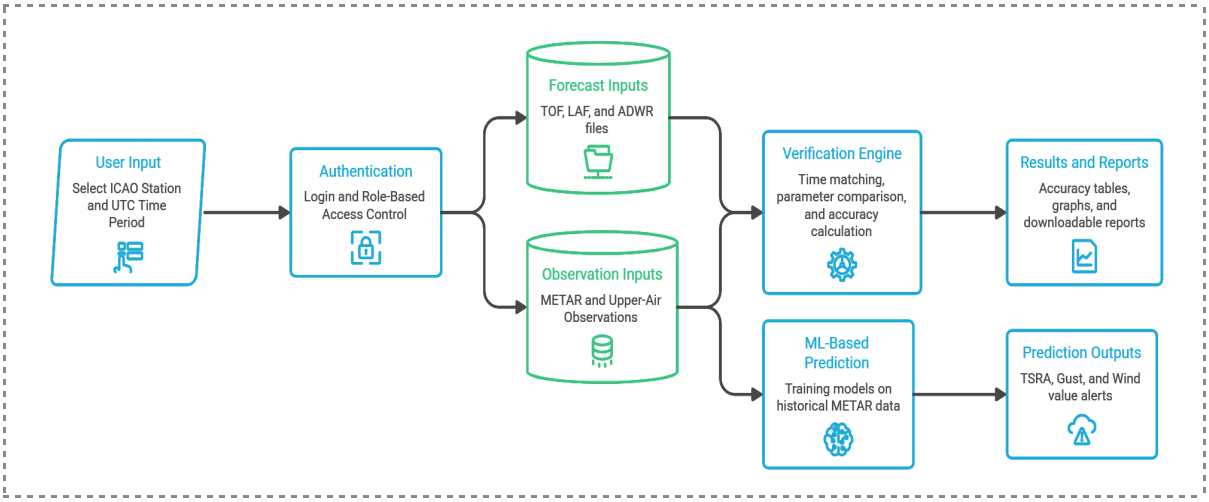


Fig. 5.1: System architecture of the Web Portal

5.2 User Interaction and Role-Based Access Control

Users log in through a login page, after which role-based access control (RBAC) directs them to one of three roles (Fig. 5.2): Super Admin (full system management), Admin (user management and user logs) and User (aviation verification dashboard).

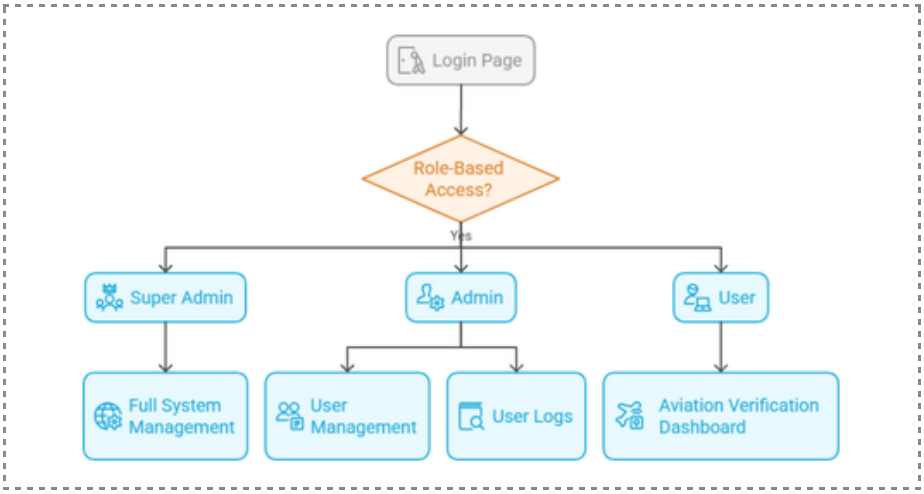


Fig. 5.2: Role-Based Access Control (RBAC) workflow

5.3 Functional Modules

The portal consists of four modules. The Take-Off Forecast module compares TOF text files with METAR observations. The Local Area Forecast module compares forecasts from LAF PDF reports with upper-air observations. The Aerodrome Warning module compares warning text files with METAR observations for thunderstorm occurrence and wind/gust exceedance. The Aerodrome Warning Prediction module uses historical METAR data to estimate TSRA and gust probabilities and wind values. Table 5.1 summarises the modules.

Table 5.1: Summary of portal modules

Module	Forecast input	Observation input	Outputs
TOF verification	TAKE-OFF text files	METAR reports	Accuracy tables, daily graphs
LAF verification	Local Area Forecast (PDF)	Upper-air observations (CSV)	Parameter-wise accuracy, summary, IMD-format report
ADWR verification	Aerodrome Warning text files	METAR reports	Warning-wise accuracy, tables, summary cards, graphs
Aerodrome Warning Prediction	Historical METAR observations	METAR (next observation)	TSRA probability, gust probability, wind estimates

The prediction module contains two logistic regression classifiers (TSRA in the next METAR; gust $\geq$ 25 kt) and one linear regression model (wind speed, gust, direction), trained on chronologically ordered METAR-derived data.

5.4 Data Flow

Forecast files and observations enter the verification path, which outputs accuracy results. Historical METAR observations enter the ML path, which outputs probabilities and wind estimates. METAR is common to both: it is the ground truth for TOF and ADWR verification and the training data for the ML module.

5.5 Model Training, Calibration and Testing Workflow

The dataset is divided chronologically into 70% training, 15% calibration and 15% testing data. The models are fitted on the training set; the alert threshold is chosen on the calibration set by maximising the F1 score; and final performance is reported on the untouched test set (Fig. 5.4). Chronological splitting is used because observations are time-dependent and random shuffling would let future information leak into training.

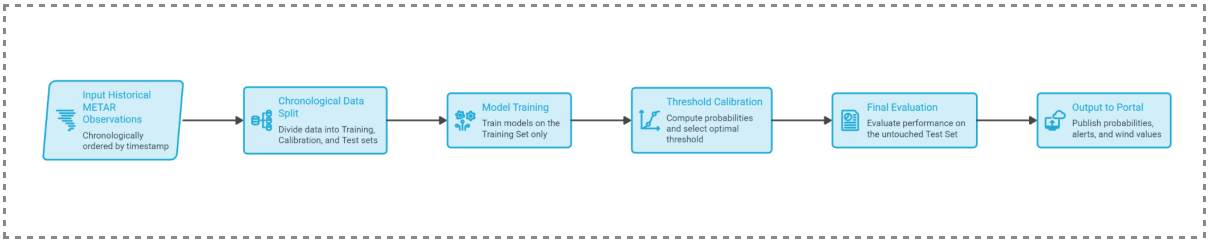


Fig. 5.4: Model training, calibration and testing workflow

5.6 Integration of Verification and Prediction

Both functions form one workflow. Verification tells forecasters how reliable past TOF, LAF and ADWR products were; prediction offers an estimate of what may happen in the next observation. Verification results can reveal weak areas (for example the lower gust/wind warning accuracy in the sample result), and the ML module targets exactly these hazards, thunderstorm and gusts. Both use METAR data, so they share the same data foundation.

6. Flow Chart

6.1 Overall Portal Workflow

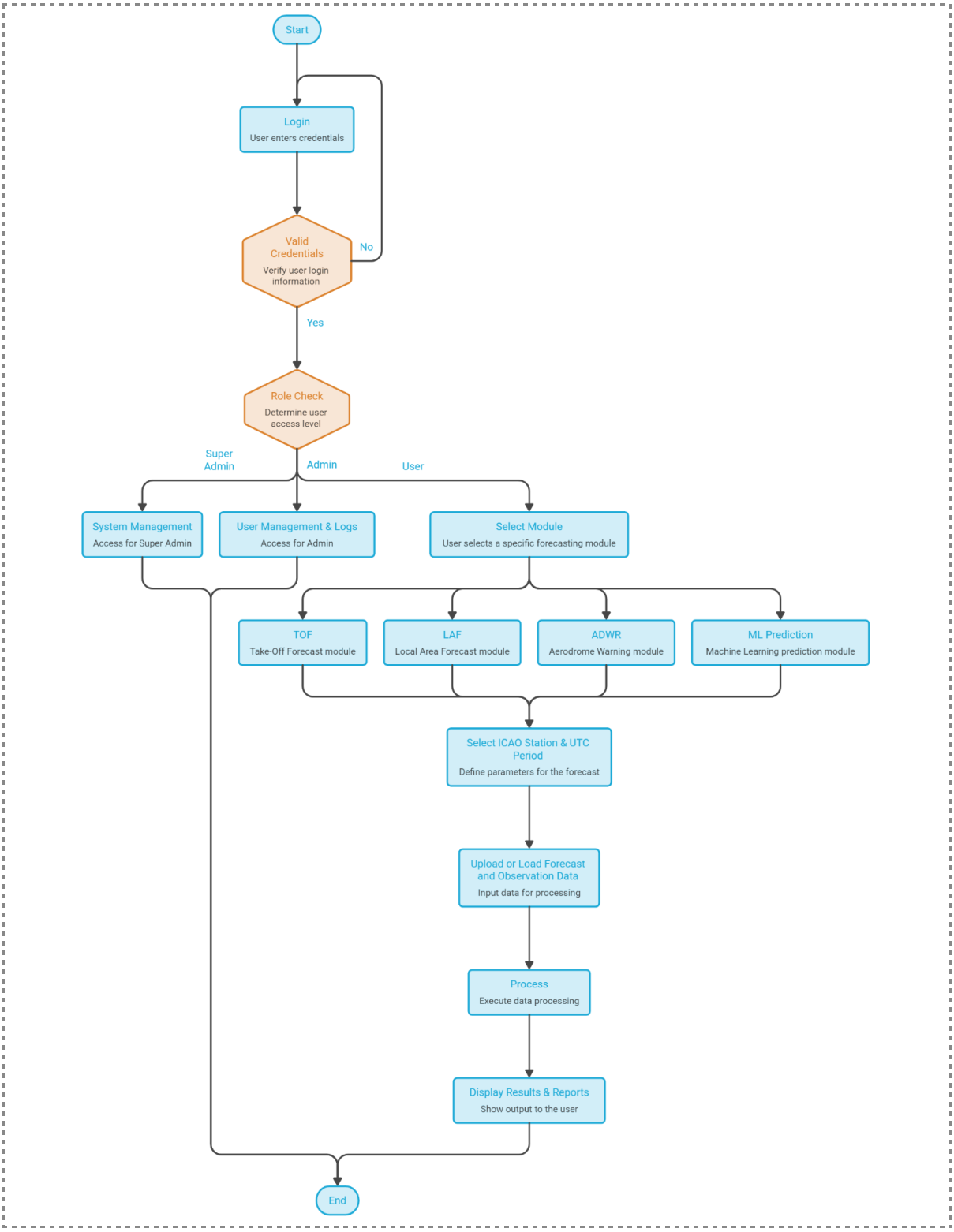


Fig. 6.1: Overall portal workflow

6.2 Forecast Verification Workflow

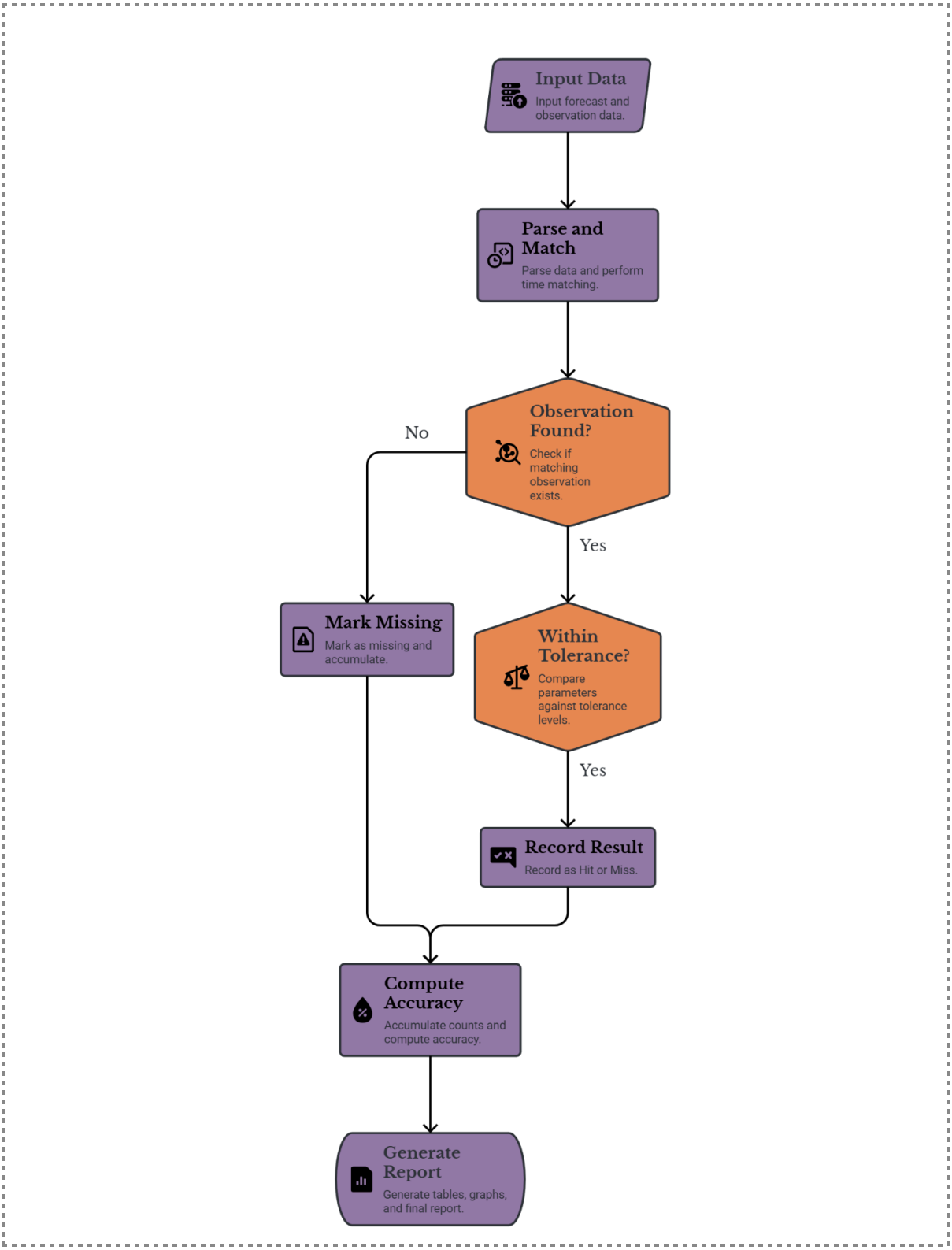


Fig. 6.2: Forecast verification workflow

6.3 ML Model Training and Prediction Workflow

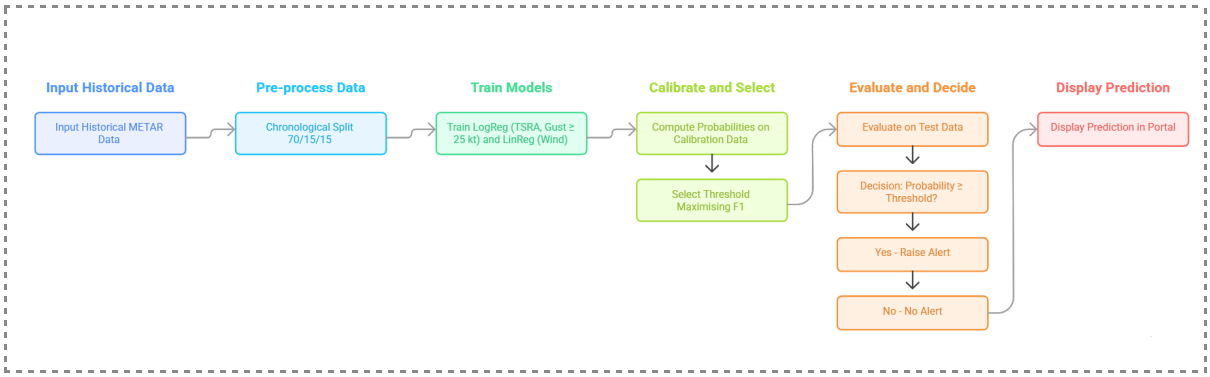


Fig. 6.3: ML model training and prediction workflow

7. System Implementation

7.1 Web Application Architecture

The portal is a web application hosted on IMD infrastructure, with a login page, role-based dashboards and a verification dashboard (Chapter 5). The frontend is built with HTML, CSS and JavaScript; the backend uses Flask (Python); the ML models are implemented in scikit-learn; and structured data is stored in SQLite. The technology stack is summarised in Table 7.1.

Table 7.1: Implementation technology stack

Component	Technology
Frontend	HTML, CSS, JavaScript
Backend	Flask (Python)
Database	SQLite
ML library	scikit-learn
ML models	Logistic Regression, Linear Regression
Hosting	IMD infrastructure
Deployment status	Testing and validation phase

7.2 Forecast and Observation Data Processing

The user selects an ICAO station and UTC period. Inputs are TAKE-OFF text files and METAR reports (TOF), Local Area Forecast PDFs and upper-air observations in CSV form (LAF), and Aerodrome Warning text files with METAR reports (ADWR). Sample inputs are shown in Figs. 7.1 to 7.3.

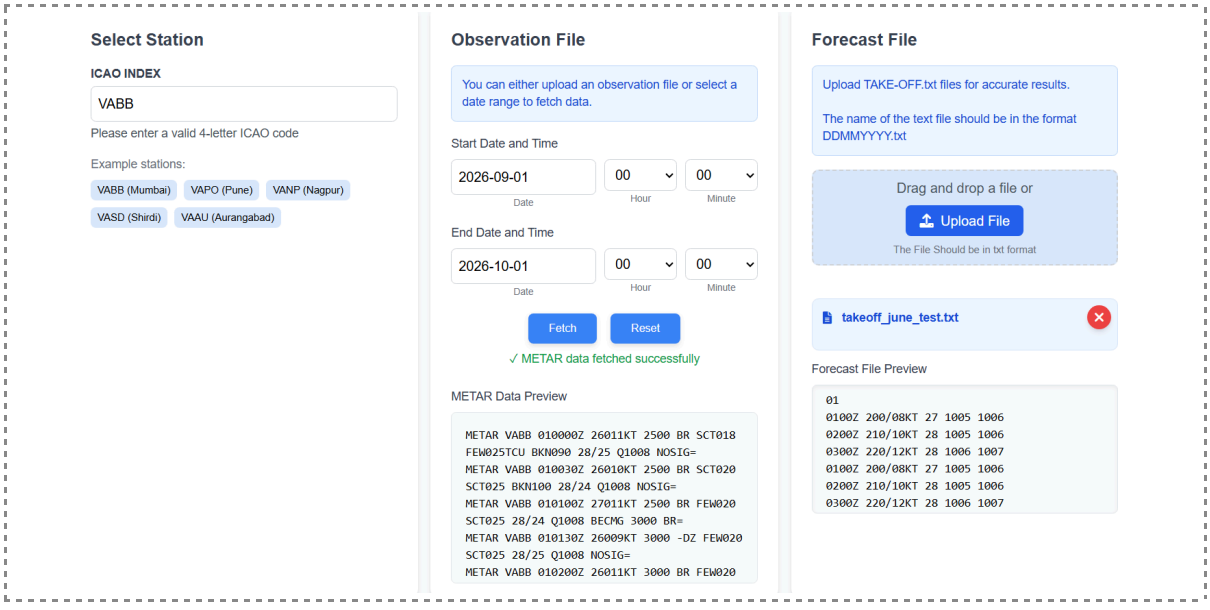


Fig. 7.1: Sample TOF text and METAR observation inputs

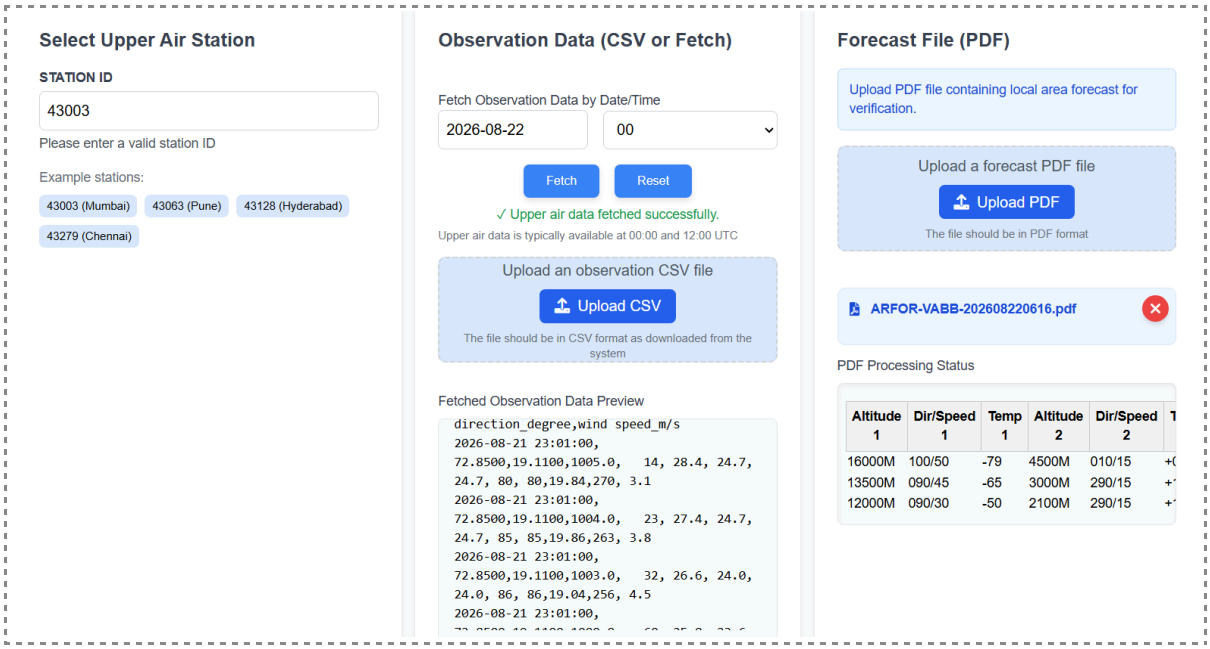


Fig. 7.2: Sample LAF report and upper-air observation inputs

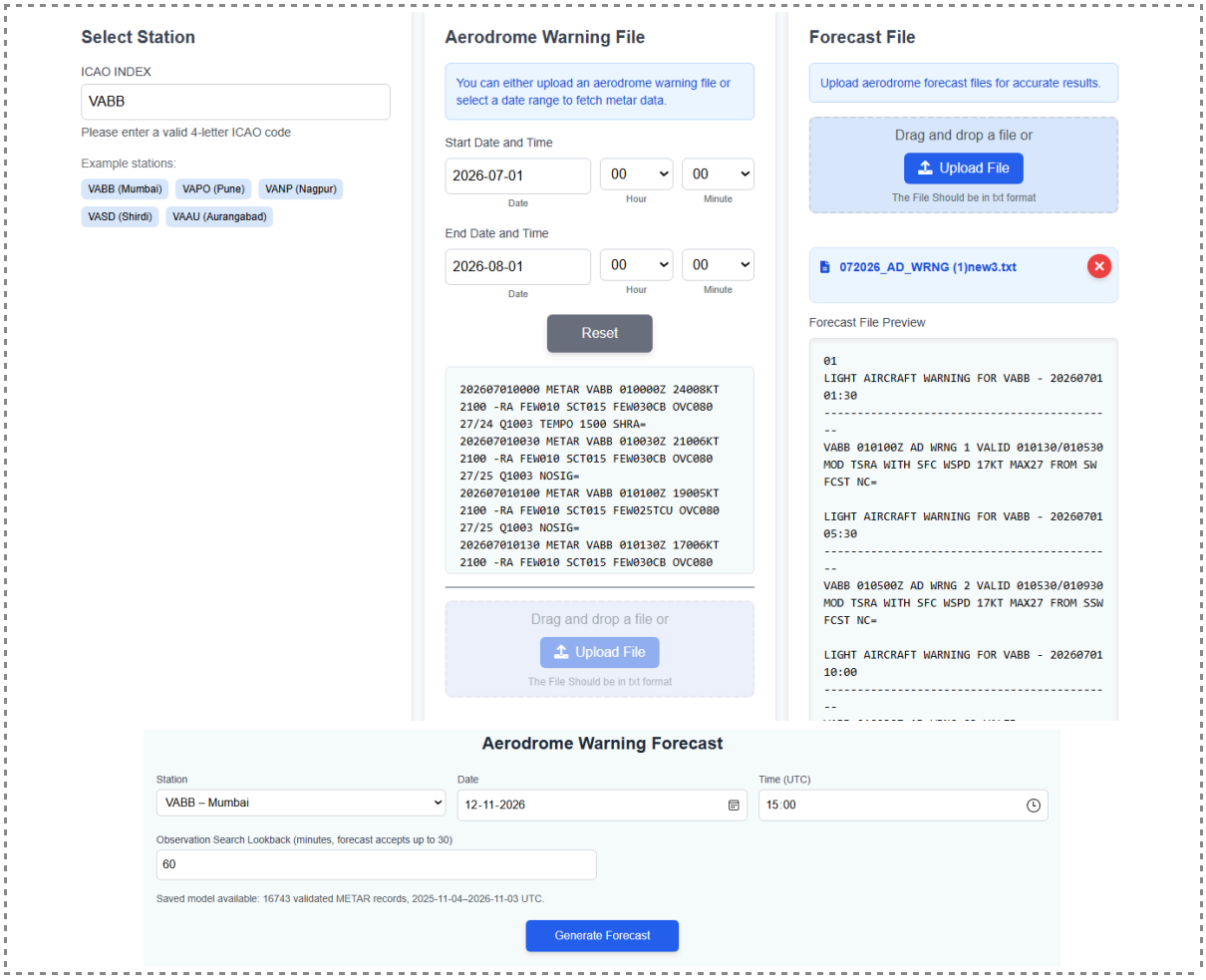


Fig. 7.3: Sample Aerodrome Warning and METAR inputs

7.3 METAR Parsing and Comparison

Each METAR line contains the observation time, station identifier, wind direction and speed (with optional gust group such as G20KT), visibility, present weather, cloud groups, temperature/dew point and QNH (for example Q1005). The Verification Engine extracts the required fields, matches each forecast time to the corresponding observation, and compares each parameter against its tolerance. For upper-air data, wind speed is given in m/s in the sounding table, so it must be expressed in knots (1 m/s ≈ 1.944 kt) before comparison with the knot-based criterion.

7.4 TOF Verification Criteria

Table 7.2: Take-Off Forecast verification criteria

Parameter	Tolerance
Wind direction	±30°
Wind speed	±5 knots
Temperature	±1 °C
QNH	±1 hPa

7.5 LAF Verification Criteria

Table 7.3: Local Area Forecast verification criteria

Parameter	Criterion
Temperature	±2 °C
Wind speed	±10 knots
Wind direction	±30°
Weather	Occurrence / non-occurrence

7.6 Aerodrome Warning Verification

Thunderstorm warnings are verified by checking for occurrences of thunderstorm weather such as TS or TSRA in the METAR observations during the warning validity. Gust/wind warnings are verified against the 25 kt threshold used consistently across the portal.

7.7 Data Preprocessing for Prediction and Chronological Splitting

Historical METAR observations are ordered by time and split into the first 70% (training), the next 15% (calibration) and the final 15% (testing). The test set contains 56,482 samples, which implies a dataset of roughly 1,66,500 observations in total.

7.8 Logistic Regression Models

For a feature vector x, the probability of the event is $P(y=1|x) = 1 / (1 + e^{-(w \cdot x + b)})$, where w and b are learned weights. Two separate classifiers are used: TSRA in the next METAR observation, and gust $\geq$ 25 knots. A probability is converted into an alert when it exceeds the chosen threshold.

7.9 Linear Regression Model

A separate linear model estimates the continuous values $\hat{y} = w \cdot x + b$ for wind speed, wind gust and wind direction.

7.10 Calibration-Based Threshold Selection

Rather than using the default 0.5 cut-off, the alert threshold is chosen on the calibration set as the value that maximises the F1 score, $F1 = 2 \cdot Precision \cdot Recall / (Precision + Recall)$. This balances missed events against false alarms, which is important when events such as TSRA are rare.

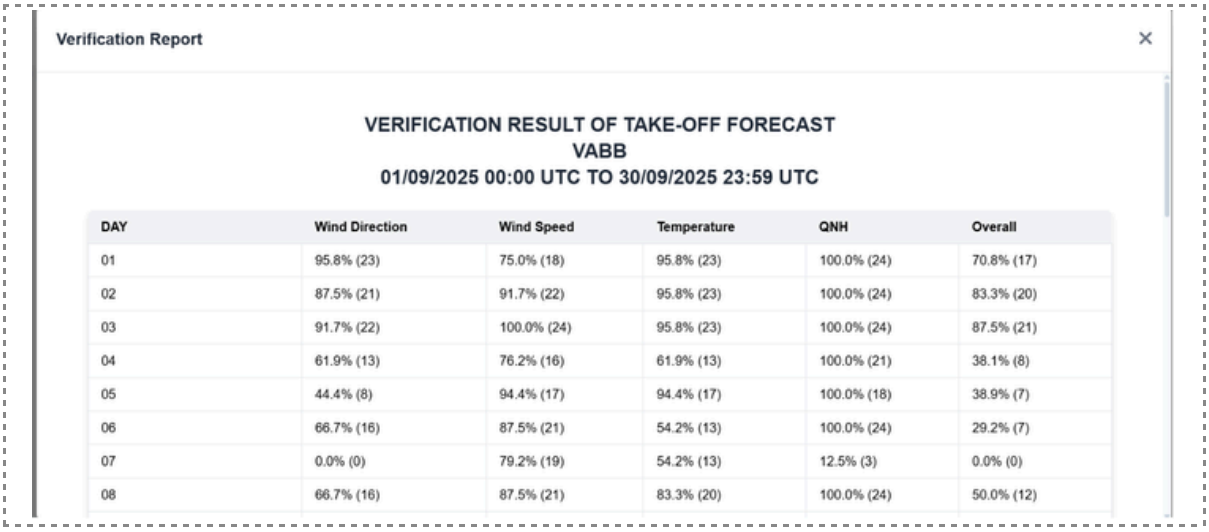
7.11 Prediction Output and Portal Integration

The module outputs event probabilities, the alert decision for the selected threshold, and estimated wind values. The intended integration is to present these in the portal as decision support beside the verification dashboard.

8. Results and Analysis

8.1 Forecast Verification Results

The portal automates verification of TOF, LAF and ADWR, and produces parameter-wise and overall accuracy, tables, graphs and reports, reducing manual effort. The results below are taken from the portal.



DAY	Wind Direction	Wind Speed	Temperature	QNH	Overall
01	95.8% (23)	75.0% (18)	95.8% (23)	100.0% (24)	70.8% (17)
02	87.5% (21)	91.7% (22)	95.8% (23)	100.0% (24)	83.3% (20)
03	91.7% (22)	100.0% (24)	95.8% (23)	100.0% (24)	87.5% (21)
04	61.9% (13)	76.2% (16)	61.9% (13)	100.0% (21)	38.1% (8)
05	44.4% (8)	94.4% (17)	94.4% (17)	100.0% (18)	38.9% (7)
06	66.7% (16)	87.5% (21)	54.2% (13)	100.0% (24)	29.2% (7)
07	0.0% (0)	79.2% (19)	54.2% (13)	12.5% (3)	0.0% (0)
08	66.7% (16)	87.5% (21)	83.3% (20)	100.0% (24)	50.0% (12)

Fig. 8.1: Results of Take-Off forecast verification, VABB, 01/09/2025 to 30/09/2025

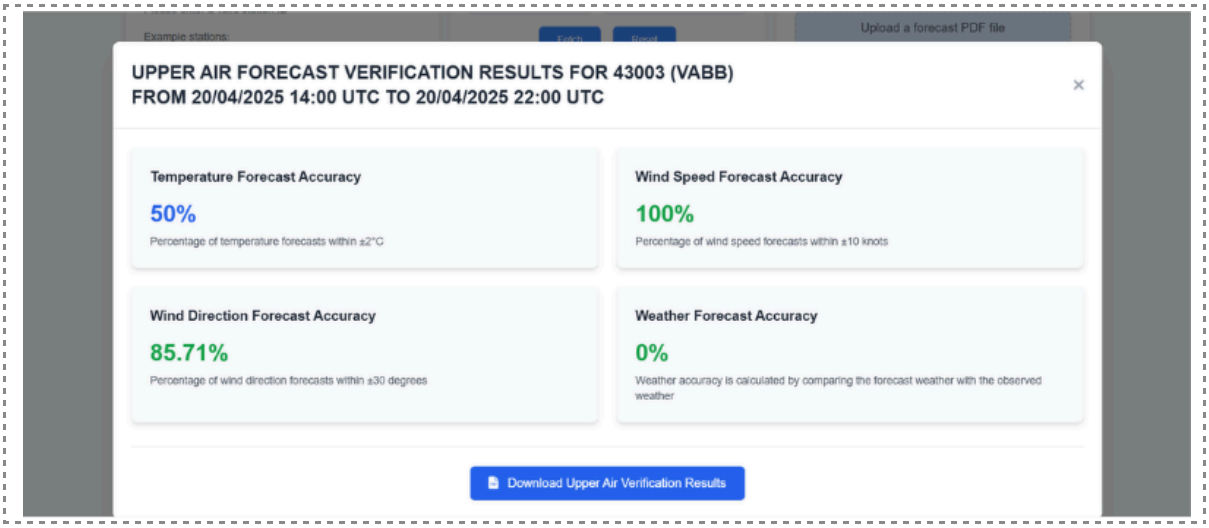


Fig. 8.2: Results of Local Area forecast verification, station 43003 (VABB), 20/04/2025 14:00–22:00 UTC

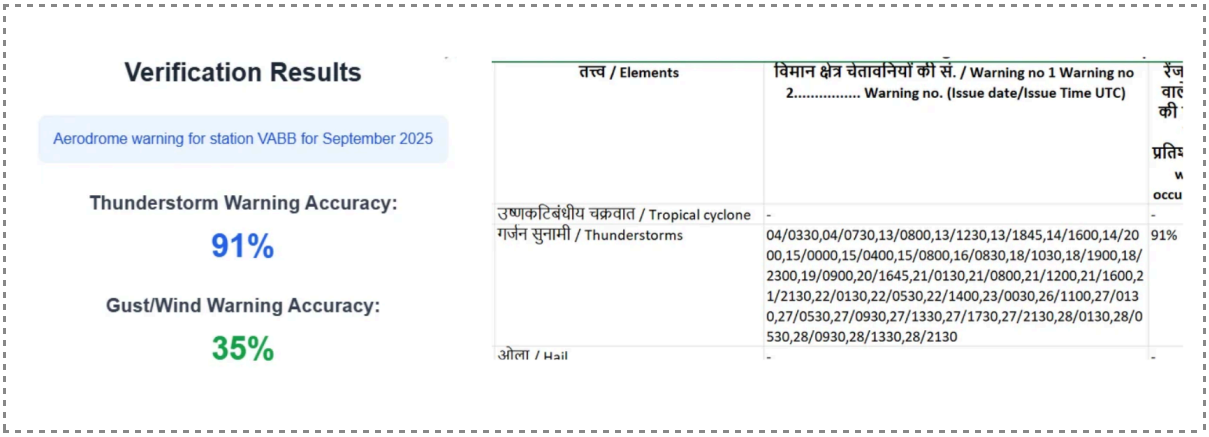


Fig. 8.3: Results of Aerodrome Warning verification

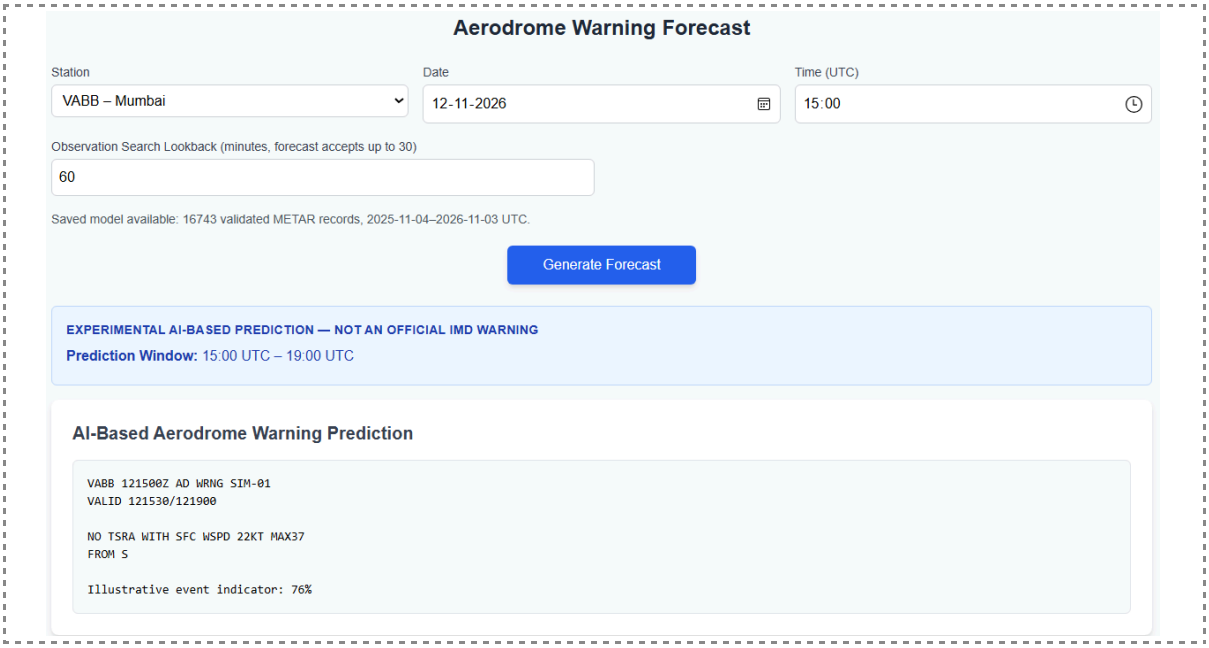


Fig. 8.4: Predicted Aerodrome Warning result

8.2 Aerodrome Warning Prediction Results

Table 8.4: Reported prediction results (currently saved model)

Metric	Value
TSRA classification accuracy	87.3%
Gust (≥ 25 kt) classification accuracy	79.6%
Test samples	56,482
Precision	77%

Metric	Value
Recall	80%
F1 score	78.5

8.3 Linear Regression Evaluation

Table 8.5: Linear regression metrics on the test set

Parameter	R^2	MAE	RMSE
Wind speed	0.68	0.18	0.24
Wind gust	0.46	0.27	0.35
Wind direction	0.34	0.21	0.29

8.4 Significance and Limitations

The reported classifiers perform well above chance on a chronologically separated test set, indicating useful predictive signal. The TSRA classifier achieved an AUC of 0.89 and the gust classifier an AUC of 0.84. The F1-optimal thresholds selected on the calibration set were 0.41 (TSRA) and 0.47 (gust). Limitations remain: station and period coverage are limited, the system is in testing at IMD, and the ML output is a statistical estimate to be interpreted by forecasters, not an official forecast.

9. Conclusion

This project delivered a web-based Aviation Forecast Verification Portal that automates the verification of Take-Off Forecasts, Local Area Forecasts and Aerodrome Warnings against METAR and upper-air observations, using the verification criteria stated in Chapter 7. It produces parameter-wise and overall accuracy, tables, graphs and reports, reducing manual effort and ensuring consistent evaluation.

The portal was extended with an ML-based Aerodrome Warning Prediction module: two logistic regression classifiers for TSRA and gust events of 25 knots or more, and a linear regression model for wind speed, gust and direction. With a chronological 70/15/15 split and F1-based threshold calibration, the saved model reports 87.3% TSRA accuracy and 79.6% gust accuracy on 56,482 test samples. The TSRA classifier achieved an AUC of 0.89 and the gust classifier an AUC of 0.84, confirming good discriminative ability.

Future extensions include expanding evaluation across additional stations and seasons, incorporating richer predictors, and adding automated verification of the ML predictions themselves through the portal. The portal is intended to support meteorological personnel as a decision-support tool and must not be treated as an autonomous replacement for official forecasting and warning procedures.

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